Intensities in wave optics

Y21 (2021) — English translation

Part A. Hole

A plane monochromatic wave with wavelength λ and intensity I_0 is incident on a round hole of radius R_0 in an opaque plane parallel to the axis of the hole. Behind the plane there is a screen at a distance L. Consider $\frac{\sqrt{\lambda L}}{R_0} \gtrsim 1$.

A1	Estimate the radius of the spot R on the screen.	**0.20**

A2	Estimate the intensity I at the center of the spot.	**0.20**

A3	Estimate the minimum achievable spot radius R_{min} when varying the hole radius.	**0.30**

Part B: Conventional Lens

A plane monochromatic wave with wavelength λ and intensity I_0 is incident on a lens of diameter D and focal length F parallel to the axis of the lens.

B1	Estimate the diameter D_F of the spot at the focal point of the lens.	**0.20**

B2	Estimate the intensity I_F at the lens focus from energy considerations.	**0.30**

B3	Now calculate the answer for the intensity I_F at the focal point of the lens exactly. Compare with the previous point.	**1.00**

Part C. Not much of a lens anymore

A plane monochromatic wave with wavelength λ and intensity I_0 is incident on an ideal parabolic mirror with parabola parameter p and diameter D parallel to the mirror axis.

C1	Find the focal length F of the mirror.	**0.30**

C2	Estimate the intensity I at the mirror focus.	**0.30**

C3	How many times will this value change when the wavelength is halved?	**0.20**

Part D. Almost like a lens

An ideal spherical mirror with radius $R = 1m$ and diameter D was illuminated with a beam of rays with wavelength $\lambda = 800$ nm and intensity I_0 parallel to the mirror axis.

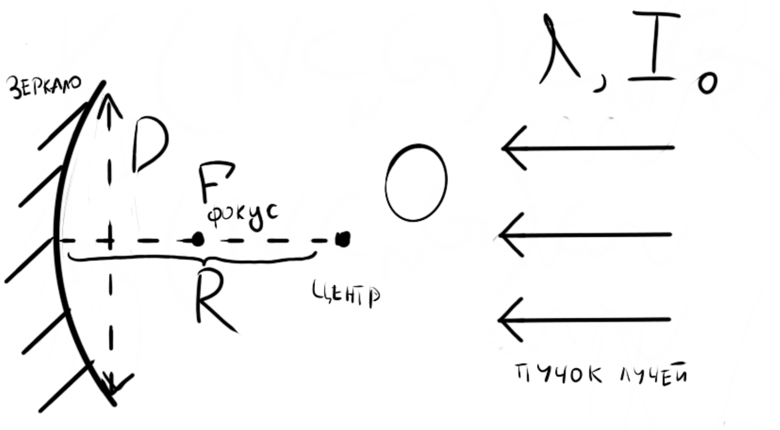

Figure 1: Figure 1.

D1	Find the focal length F of the mirror.	**0.20**

D2	Estimate the size of the spot d in the focal plane of the mirror, for two mirror diameters: $D_1 = 1\ cm$, $D_2 = 15\ cm$.	**1.30**

D3	For these two mirror diameters ($D_1 = 1\ cm$, $D_2 = 15\ cm$), estimate the intensity at the focus.	**0.80**

D4	How do the intensities at the focus found at the previous point depend on the wavelength λ?	**0.70**

Part E. Not a lens at all

A normally plane wave is incident on a diffraction grating with a period d and a slit width a. Wavelength $\lambda = d/100$, plane wave intensity I_S. Further, the intensity of the maximum should be understood as the intensity of a plane wave propagating in the direction of the maximum with number h.

E1	Determine the intensity of the zero maximum I_0. Express the answer in general terms in terms of d and a. Provide answer for $d = 3a$.	**1.00**

E2	Determine the intensity I_h of the maximum with number h. Express the answer in general terms in terms of d and a. Provide answer for $d = 3a$.	**1.00**

E3	What fraction of the energy β falls into all maxima on the screen except the zero one ($h \neq 0$)? Calculate the answer for $d = 3a$.	**1.00**

E4	What fraction of the energy γ is absorbed by the non-transmitting part of the diffraction grating? Calculate the answer for $d = 3a$.	**1.00**

Source: `https://pho.rs/p/235`